Creating Static Website

Overview

- Host a static website by using Amazon S3.
- Enable versioning in Amazon S3.
- Implement a data lifecycle strategy in Amazon S3.
- Implement a disaster recovery (DR) strategy in Amazon S3.

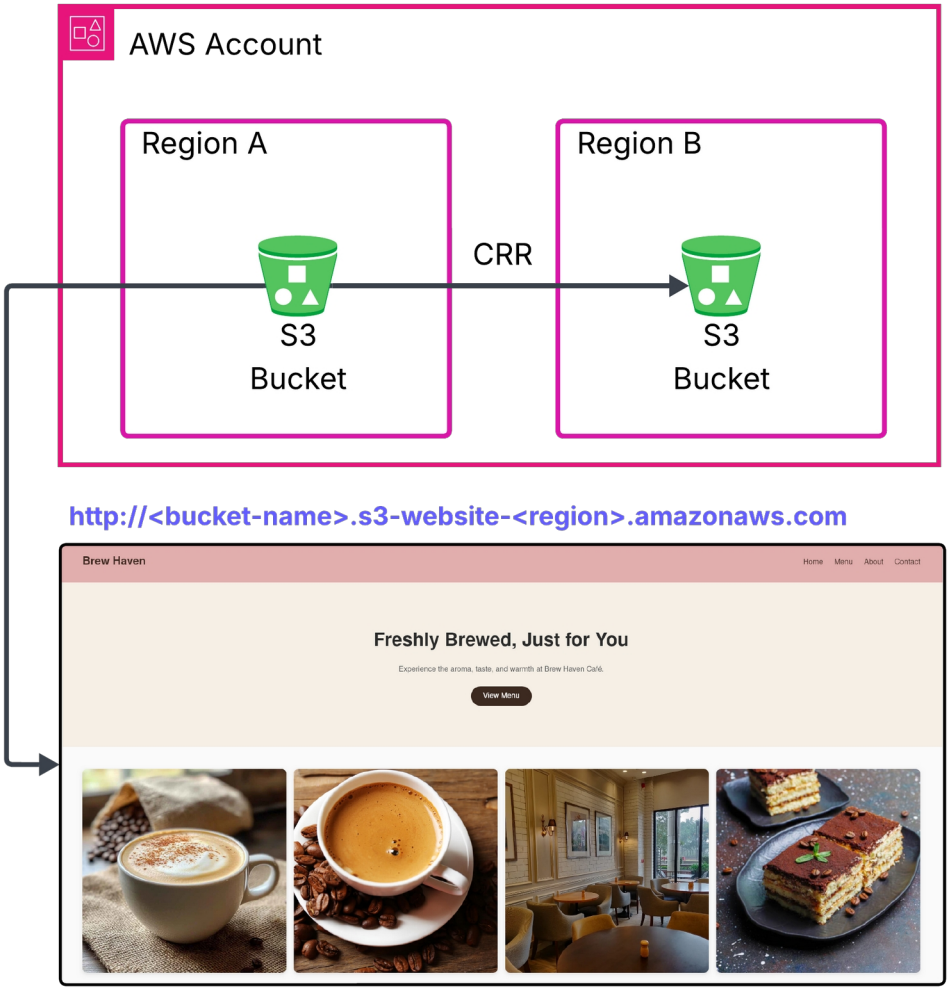


Figure 1: Static Web Site Hosting With Cross Region Replication in S3

Amazon S3 is unlimited object storage device. A bucket is a container for objects that are stored in S3. Buckets have unique name across all AWS customers account and objects have globally unique url. Objects have a key, version id, value, metadat and subresources. Single object can be of 5 TB.

Amazon S3 can be used to **host static website**, **create data lake**, **manage content**, **distribute data** etc.

Host a Static Website By Using Amazon S3

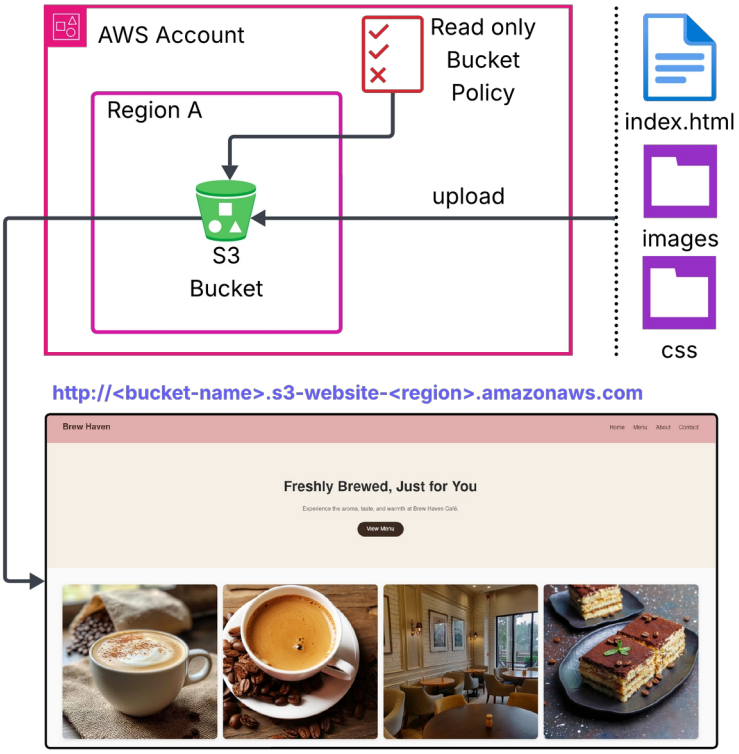
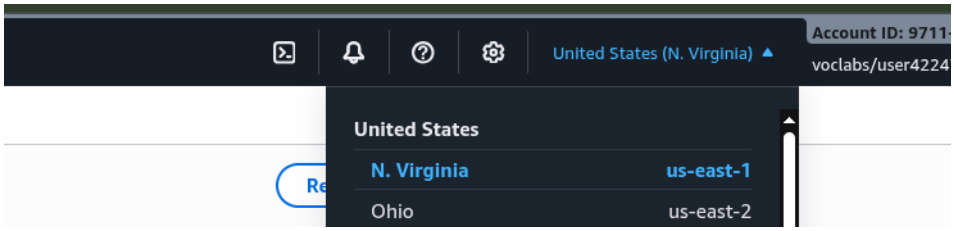


Figure 2: Hosting Static Website in Amazon S3

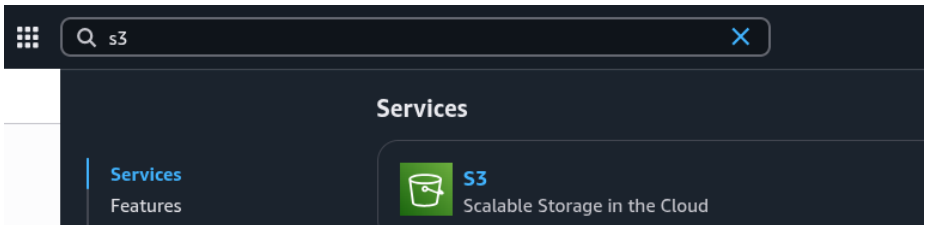
To host a static website, configure an S3 bucket for website hosting, upload website content to the bucket and attach a bucket policy that allows public read access to the objects. Get example content from here.

Step By Step Guide

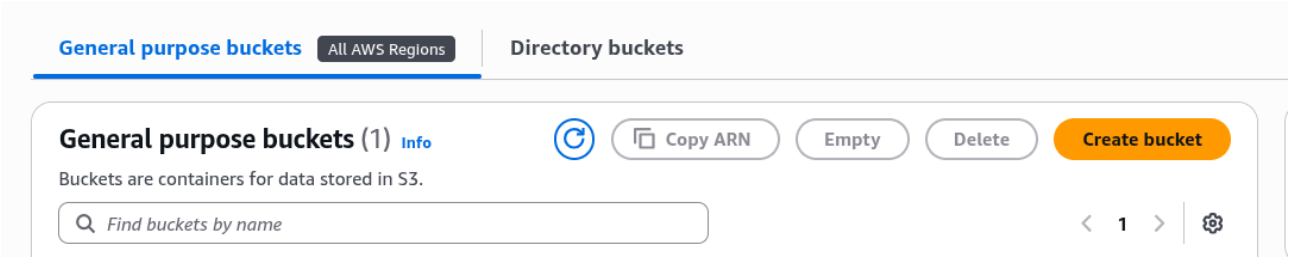
1. Login to AWS Console. Select the appropriate region.



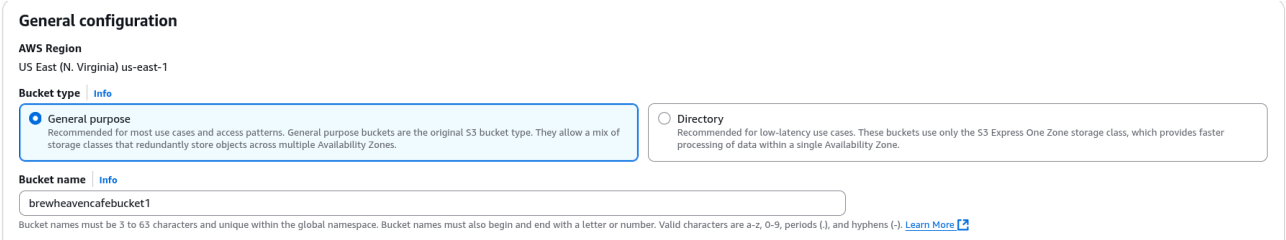
2. Search fo S3 and select it.



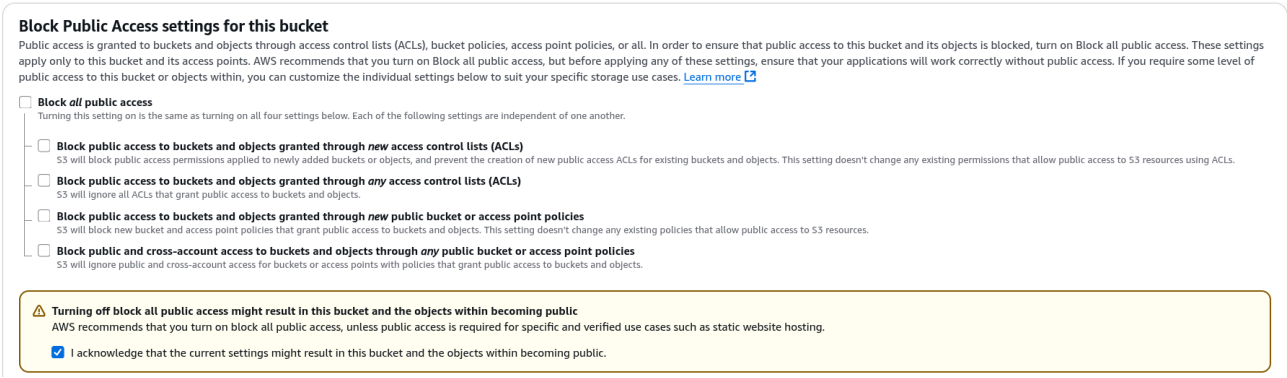
3. Click *create bucket* to create a new bucket.



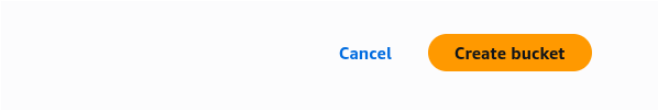
4. Give a unique name to your bucket.



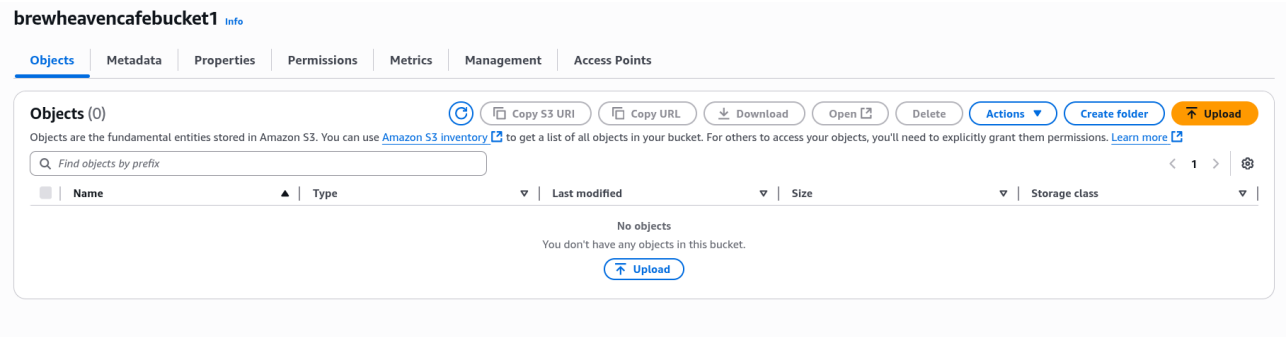
5. Clear *Block all public access*.



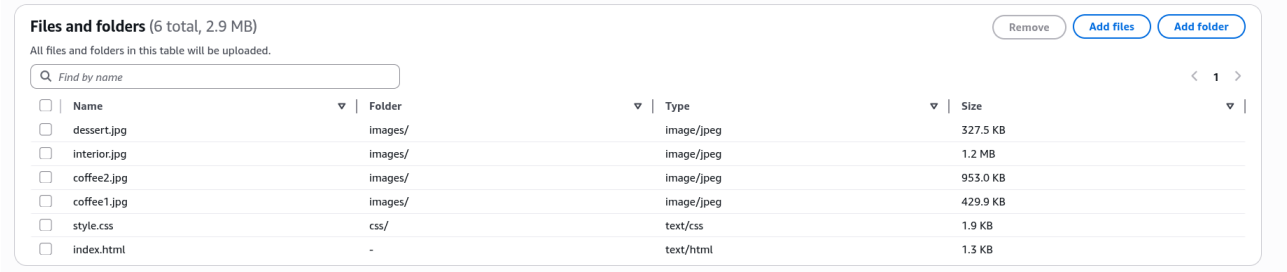
6. Keep all configurations as it is and proceed with *create bucket*.



7. After creating bucket open it and select *upload*.

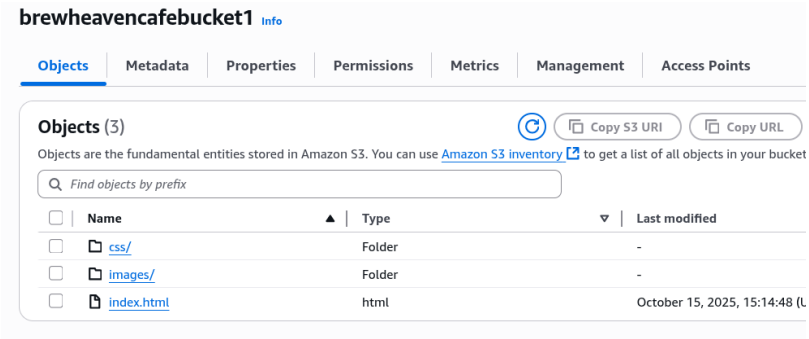


8. Select *add files* and *add folder* to add files and folder respectively.

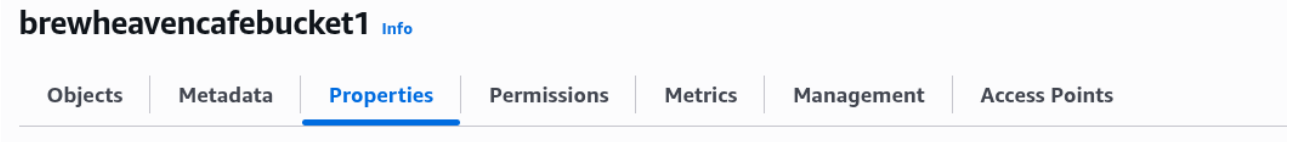


9. Att the bottom select upload to *upload* all the added files.

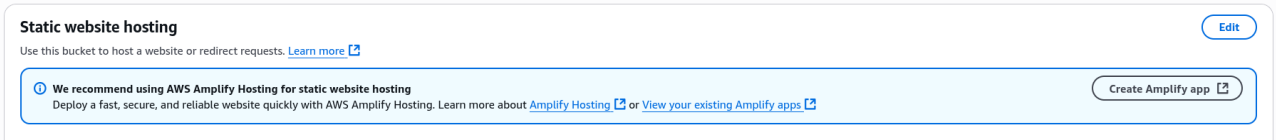
10. Close it, and return to the bucket console.



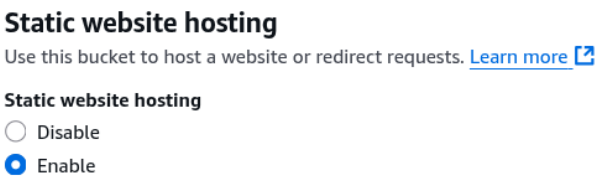
11. Select *Properties to* open it.



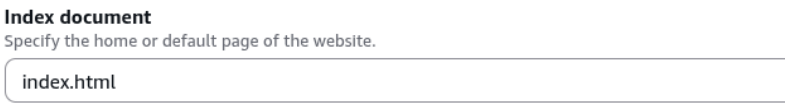
12. Scroll down to find *Static website hosting* section and click *Edit*.



13. Enable Static website hosting.

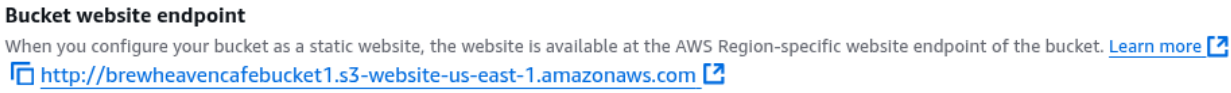


14. Add index.html in *Index document* section.



15. Proceed with *Save changes*.

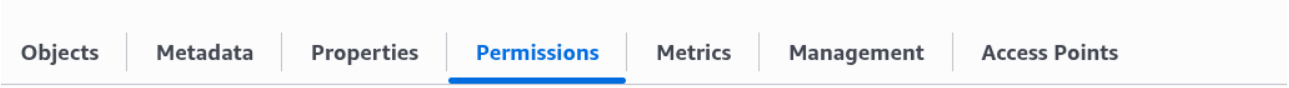
16. In *Properties* scroll down to *Static website hosting* section to get bucket website endpoint.



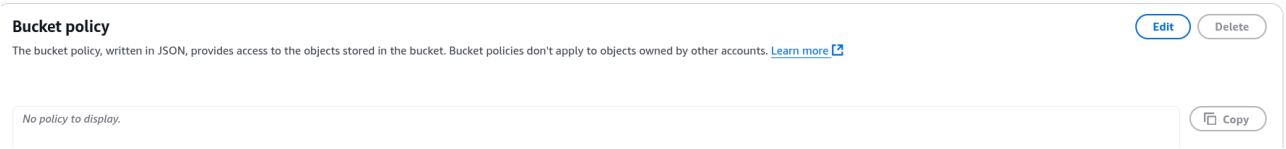
17. Open the endpoint.

Oops! Need to add bucket policy to give public read access.

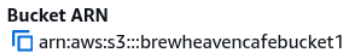
18. Return to the console and select *Permissions*.



19. In *Bucket policy* section click *Edit* to edit the bucket policy.

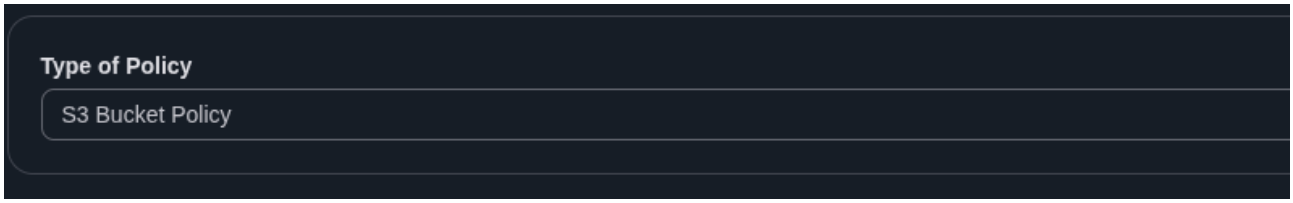


20. Copy the *Bucket ARN*.



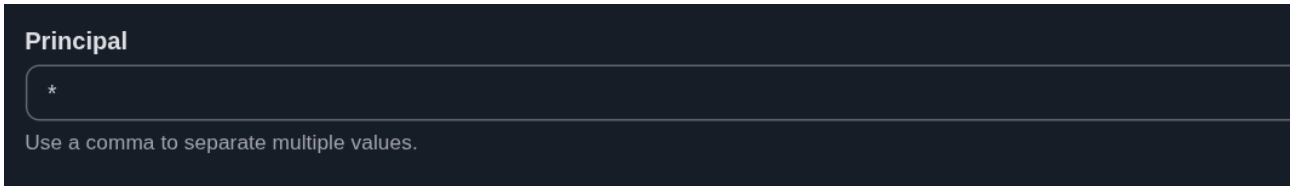
21. Open the Policy Generator.

22. In policy generator, for *Type of Policy* select *S3 Bucket Policy*.

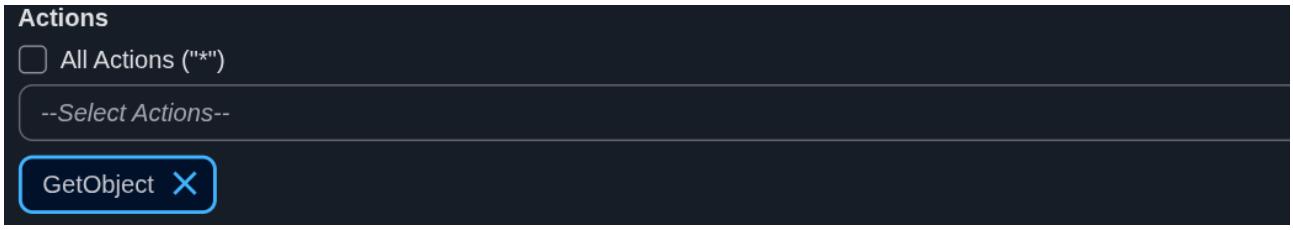


23. For *Effect* select *Allow*.

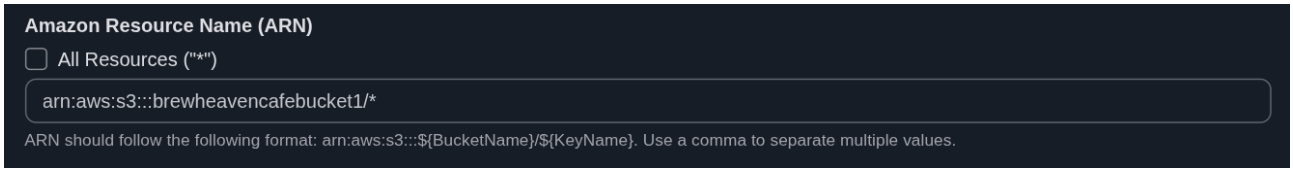
24. For *Principal* add * for everyone.



25. For *Actions* add *GetObject* only.



26. Add Bucket ARN with trailing /*.



27. Select *Add Statement*

28. Verify and Proceed with *Generate Policy*.

Statements added (1)

You added the following statements. Click the button below to Generate a policy.

Principal(s)	Effect	Action	Resource(s)	Condition(s)
*	Allow	s3:GetObject	arn:aws:s3:::brewheavencafebucket1/*	*None*

Step 3: Generate policy

A policy is a document (written in the Access Policy Language) that acts as a container for one or more statements.

Generate Policy

29. You should get the policy similar to this. Copy it.

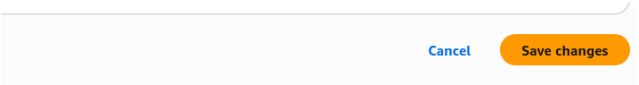
```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Sid": "Statement1",
6        "Effect": "Allow",
7        "Principal": "*",
8        "Action": [
9          "s3:GetObject"
10       ],
11       "Resource": "arn:aws:s3:::brewheavencafebucket1/*"
12     }
13   ]
14 }
```

30. Go back to bucket console and paste the policy.

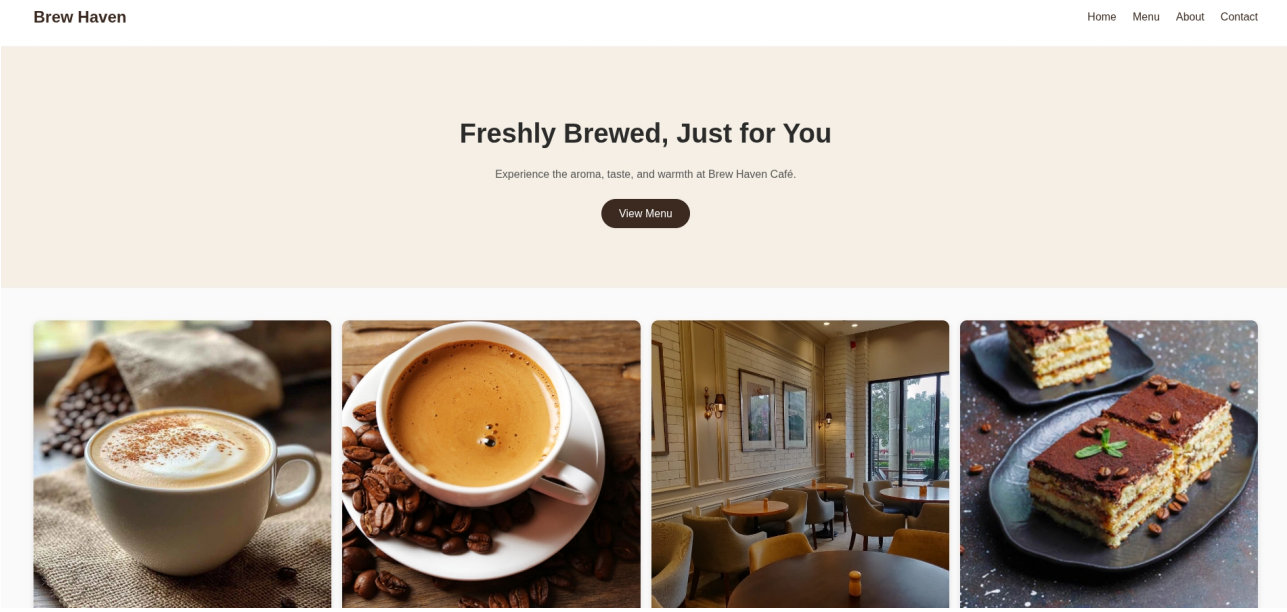
Policy

```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Sid": "Statement1",
6        "Effect": "Allow",
7        "Principal": "*",
8        "Action": [
9          "s3:GetObject"
10       ],
11       "Resource": "arn:aws:s3:::brewheavencafebucket1/*"
12     }
13   ]
14 }
```

31. Proceed with *Save changes*.



32. Go back to *Properties* and Open the endpoint.



Congratulations ! Website is live now.

Enable Versioning in Amazon S3

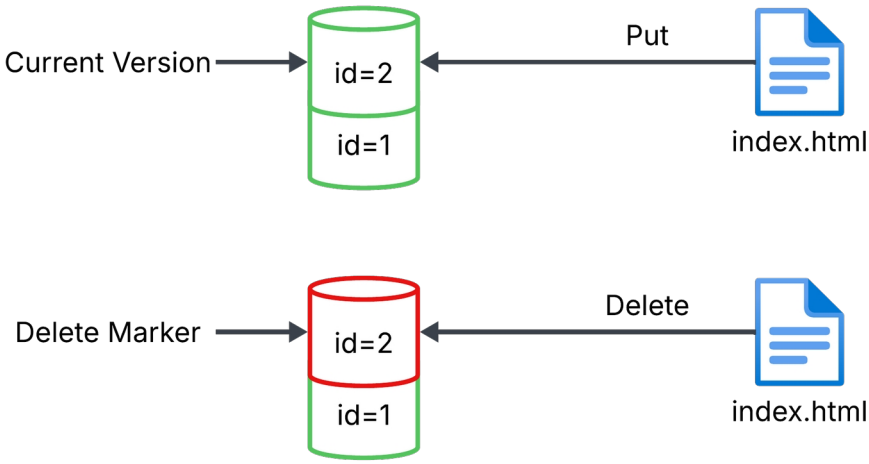


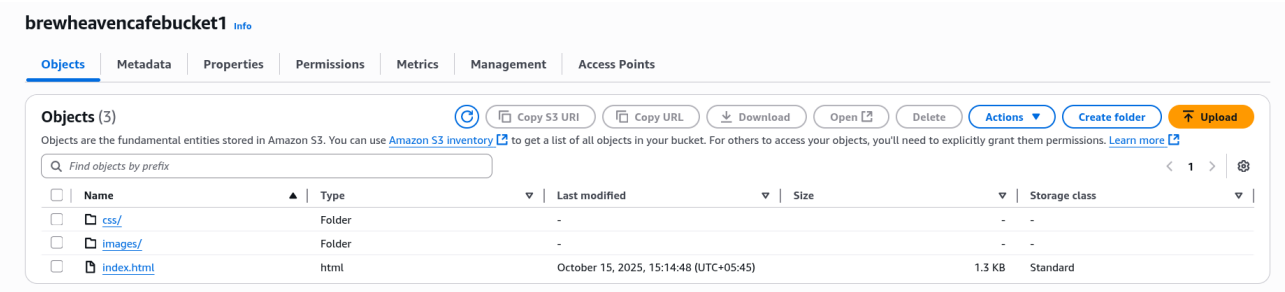
Figure 3: Versioning in S3 bucket

Enabling Versioning in one way to protect data with Amazon S3 form accidental deletion. Once enabled it can only be suspended.

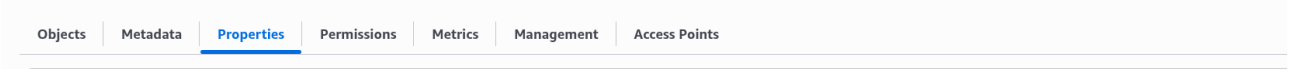
When object with same key is uploaded in version enabled bucket, it creates new object with different version ID and both are retrievable by version ID. When deleting object, a delete marker is added but the object is still retrievable by the version ID.

Step By Step Guide

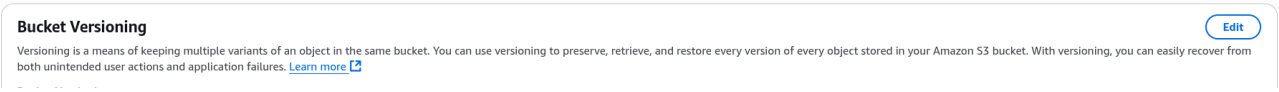
1. Open the S3 bucket console.



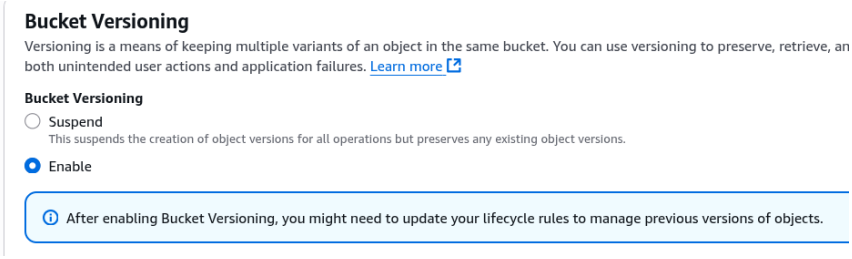
2. Select *Properties*.



3. Click *Edit* on *Bucket Versioning* to edit versioning.



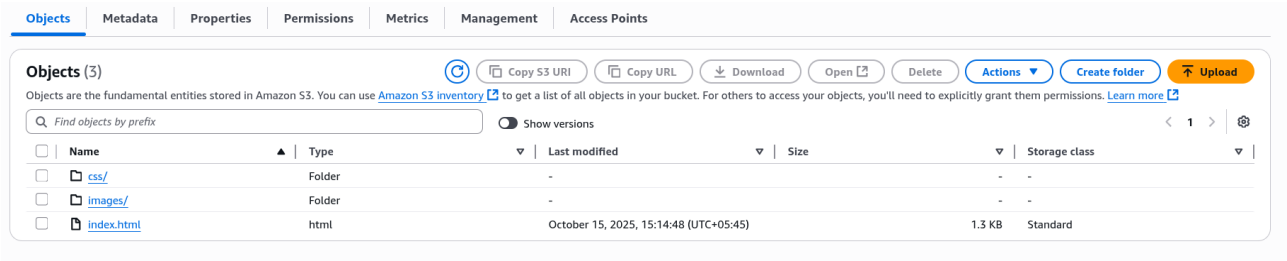
4. Enable Bucket Versioning.



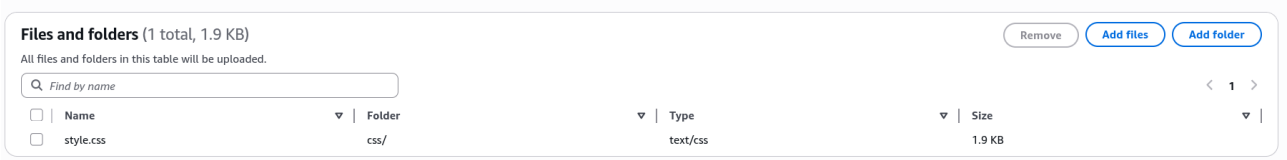
5. Proceed with save changes.

6. To test versioning, Make some changes in your css file using any code editor.

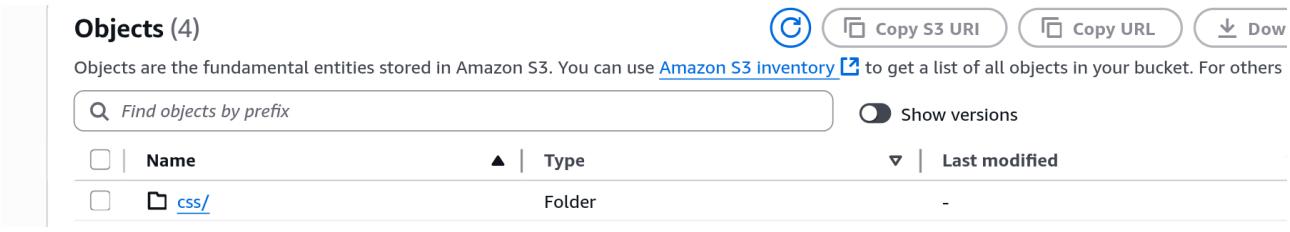
7. In the console go to *Objects*, and select *Upload*.



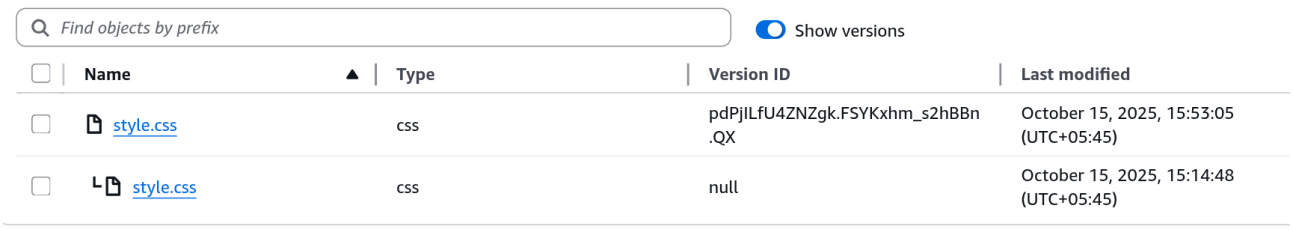
8. Add recently edited css file and upload it.



8. Close it and select *css/* prefix to open it.



9. Turn on *show versions*.



Congratulations ! There is one current version and one previous version marked with L.

Data Lifecycle Strategy in Amazon S3

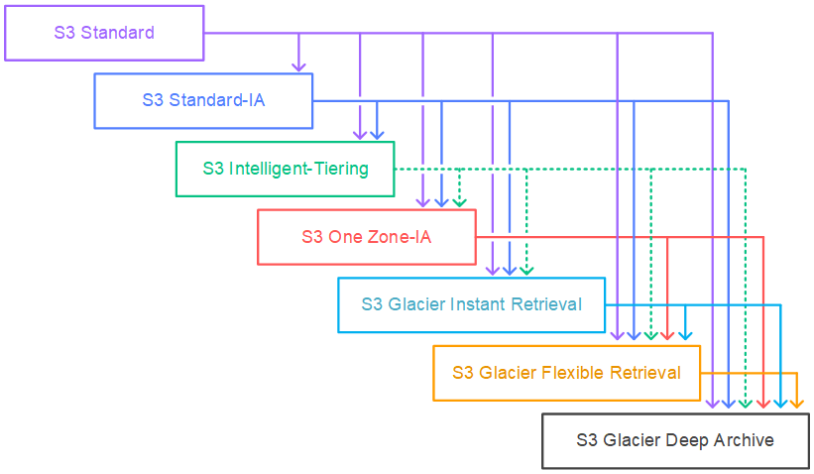


Figure 4: Data Lifecycle in Amazon S3

S3 lifecycle policy automatically transfers data to a different storage class without any changes to the website. Transition actions transition to another storage class. Expiration actions define when object expire.

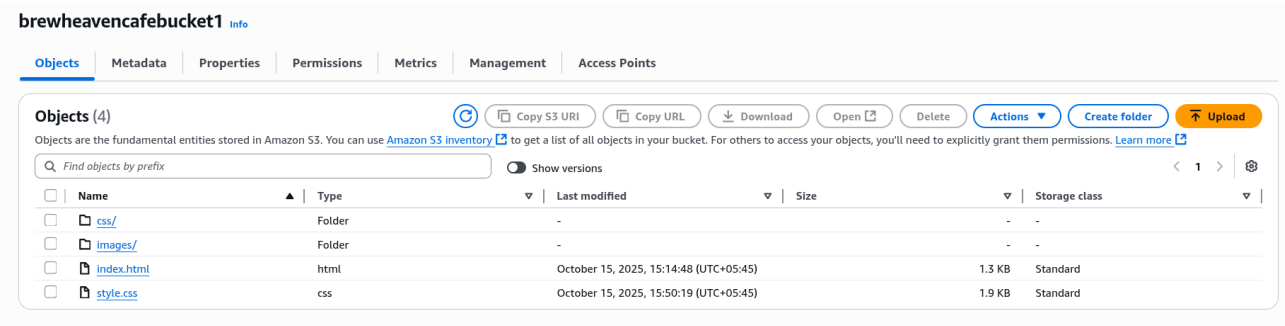
We will configure two rules:

1. Rule one, move previous versions of all source bucket objects to S3 Standard-IA after 30 days.
2. Rule two, Delete previous versions of the objects after 365 days.

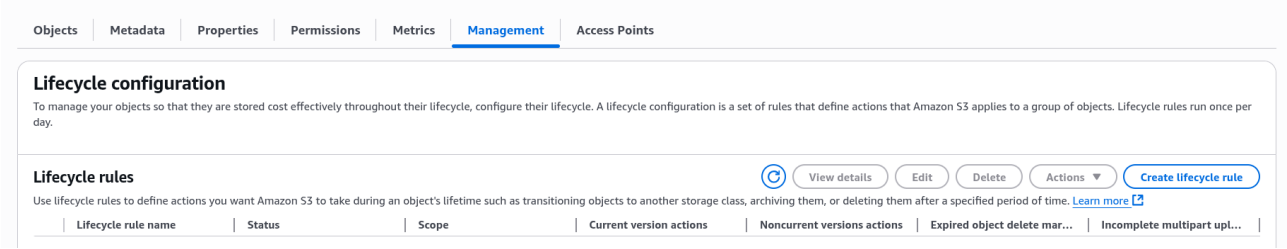
According to the Well-Architected Framework, in practice, lifecycle strategy should be based on the criticality and sensitivity of data, and legal and organizational requirements. You should consider factors such as data retention duration, data destruction, data access management, data transformation, and data sharing.

Step By Step Guide

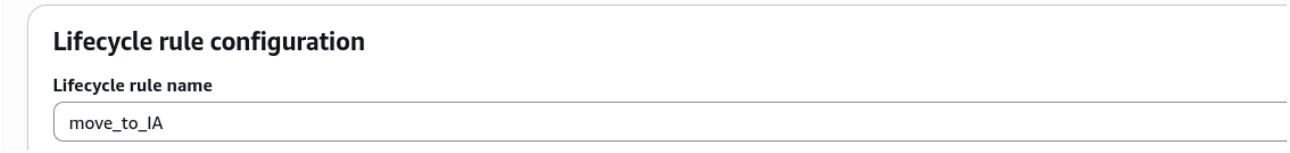
1. Open the bucket console.



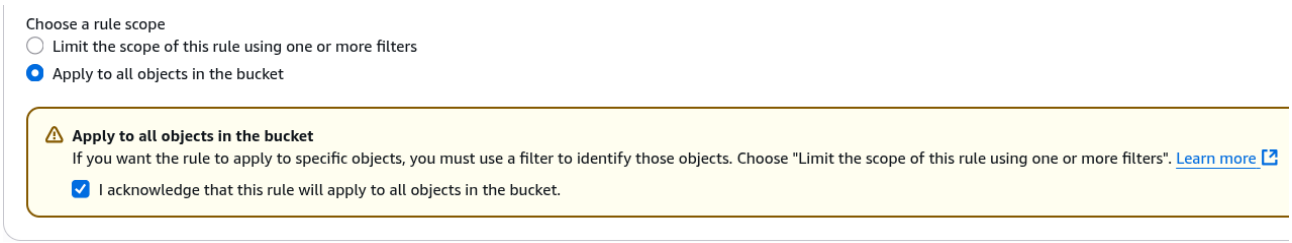
2. Select *Management* and select *Create lifecycle rule*.



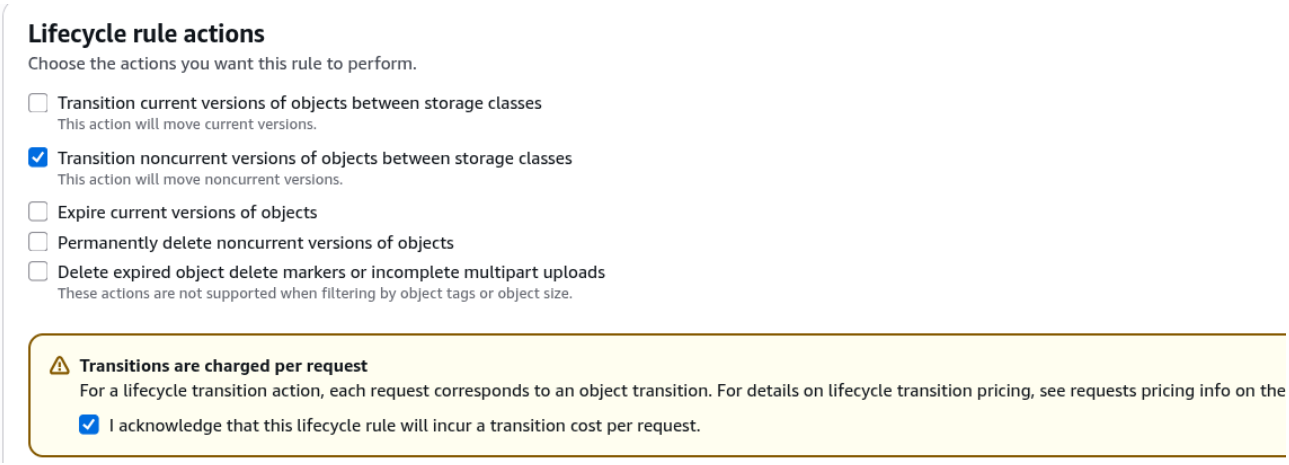
3. Name the first rule as *move_to_IA*.



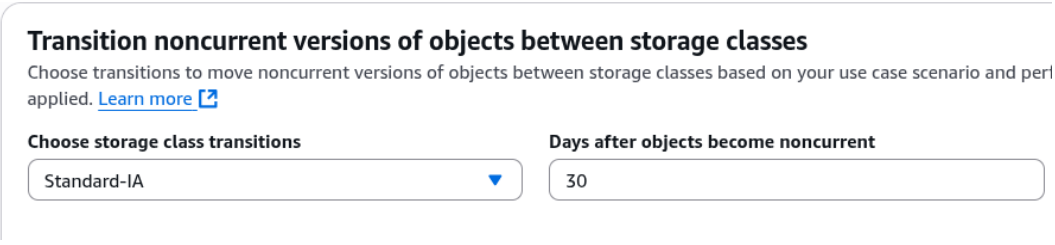
4. Choose *Apply to all objects in the bucket*.



5. Select *Transition noncurrent versions of ojbects between storage classes*.

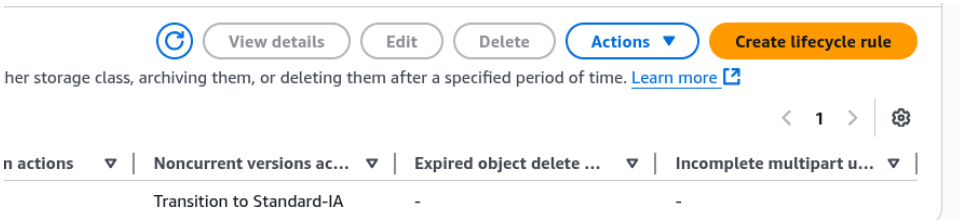


6. Select *Standard-IA* and Set number of days to *30*.

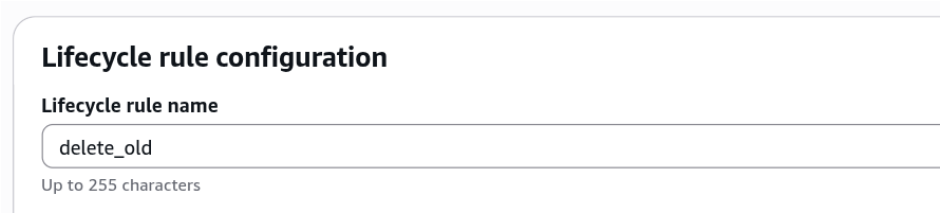


7. Proceed with *Create rule.*

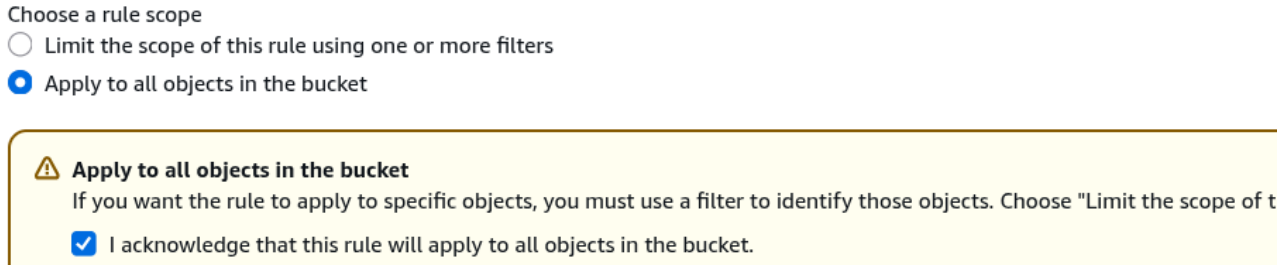
8. Select *Create lifecycle rule* to create second rule.



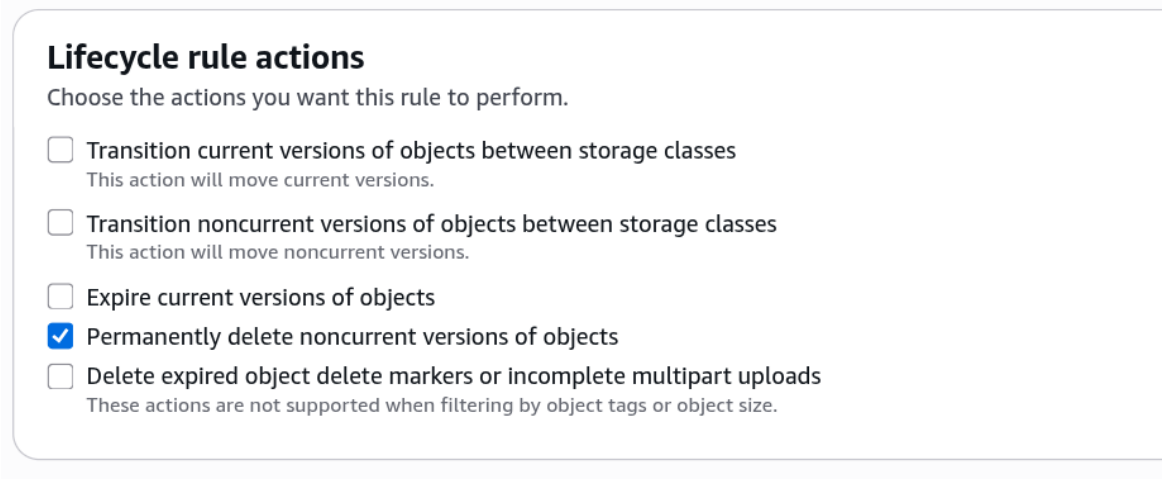
9. Name the second rule as *delete_old*



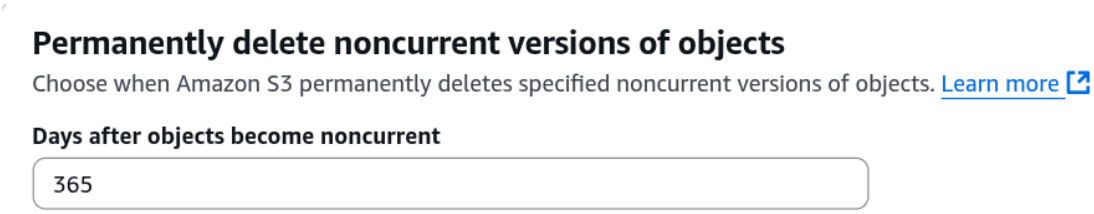
10. Choose *Apply to all objects in the bucket*.



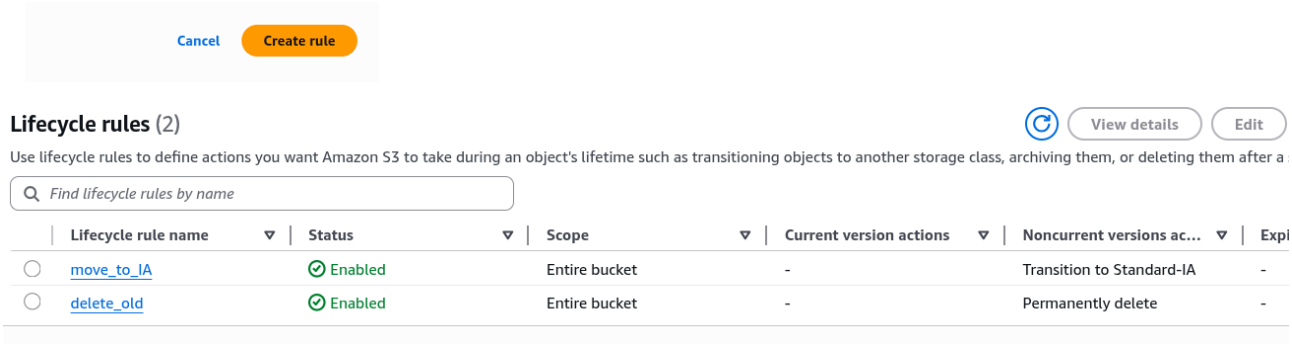
11. In Lifecycle rule actions select *Permanently delete noncurrent versions of objects.*



12. Set the Days to *365*.



13. Proceed with *Create rule*.



Congratulations ! You have two Lifecycle rules.

Disaster Recovery (DR) Strategy in Amazon S3

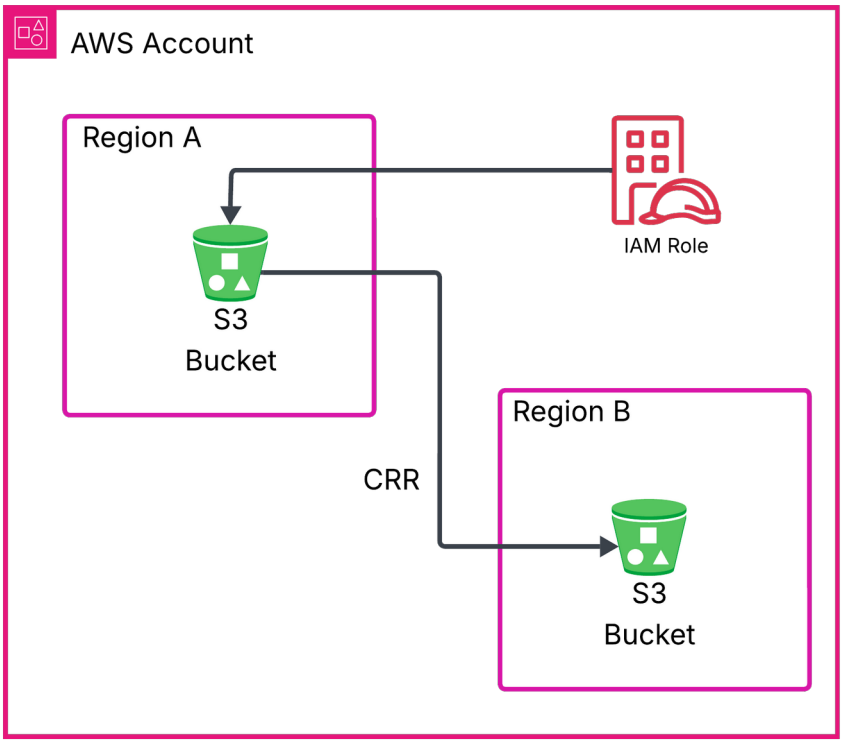
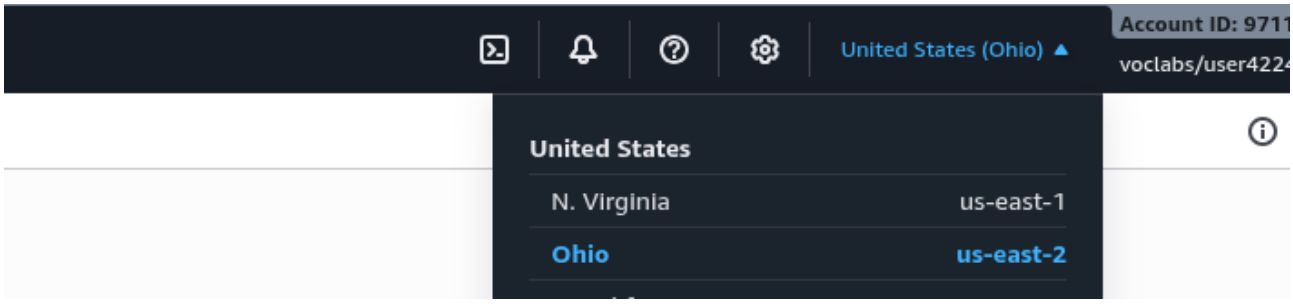


Figure 5: Cross Region Replication in Amazon S3

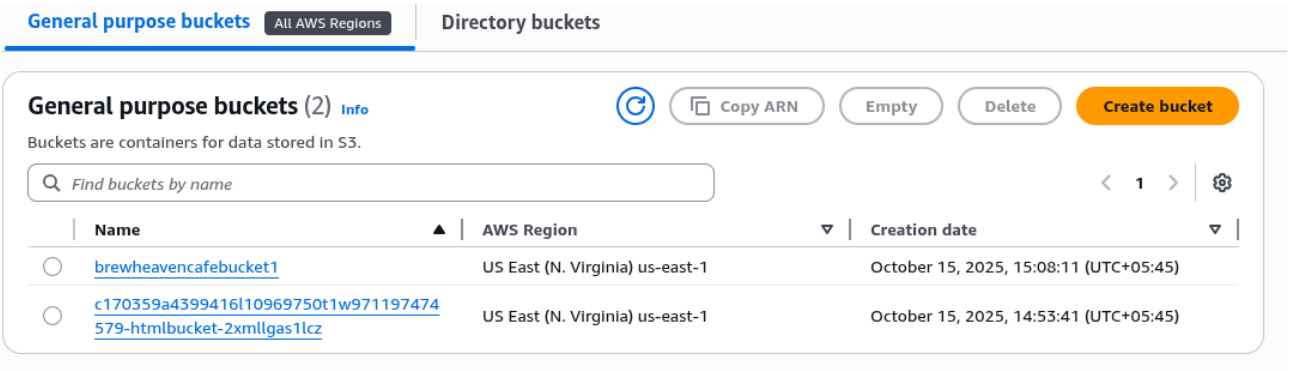
Cross-Region Replication (CRR) is another feature of Amazon S3 used to replicate, back up and archive critical data. It enhances durability and supports for Disaster Recovery (DR).

Step By Step Guide

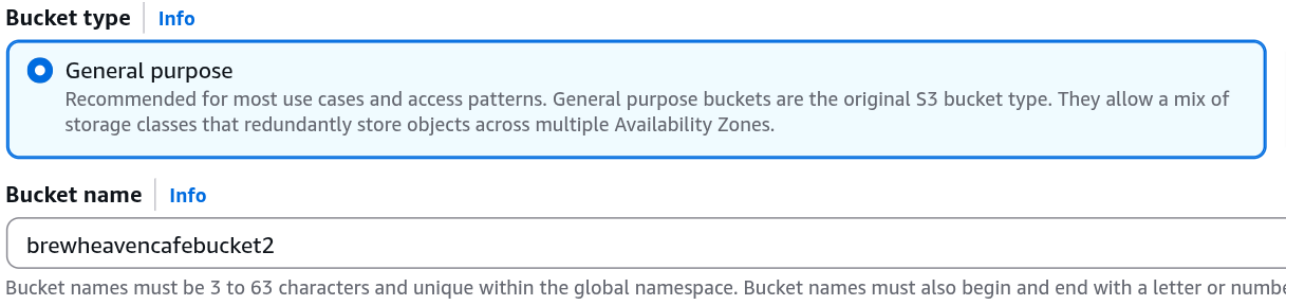
1. Select a different region to create another bucket.



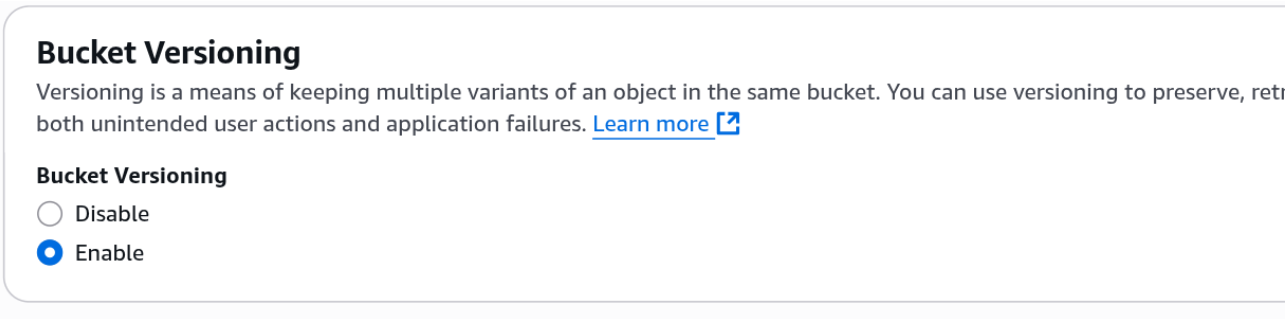
2. Open S3 Console and select *Create bucket*.



3. Give another unique name to the second bucket.

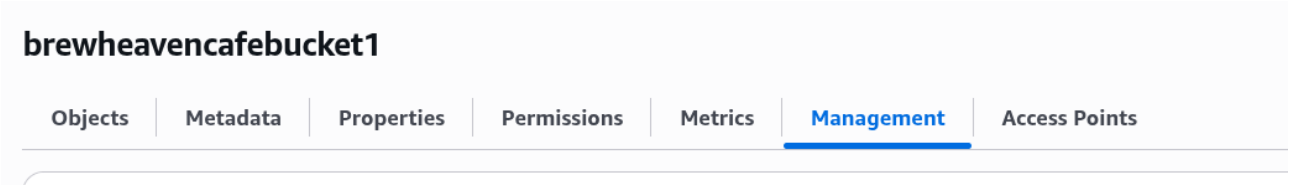


4. Enable *Bucket Versioning*.

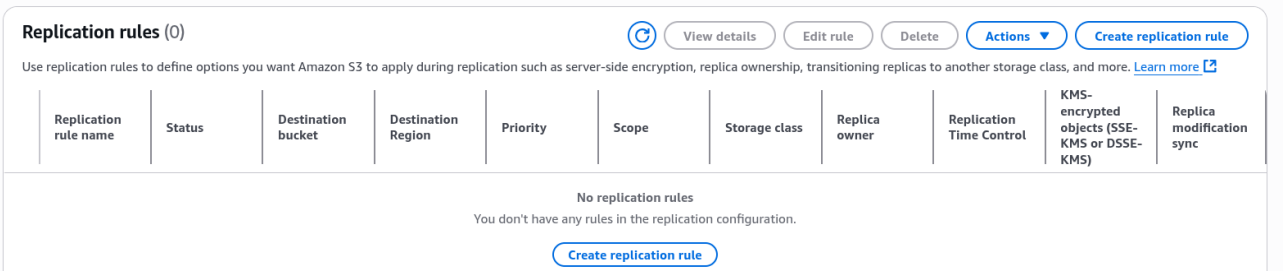


5. Keeping all the configurations default and proceed with *Create bucket*.

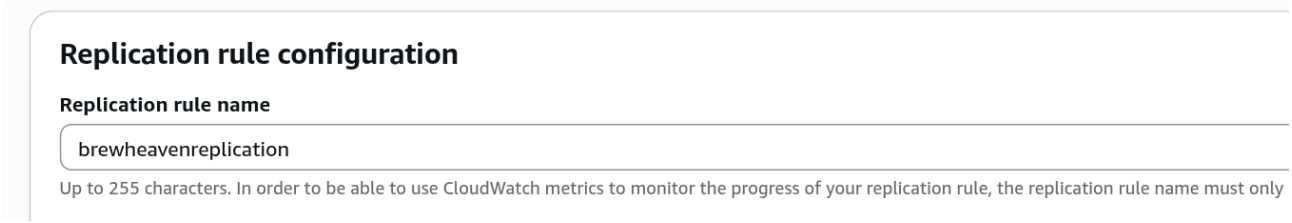
6. Open the first bucket from the console and select *Management*.



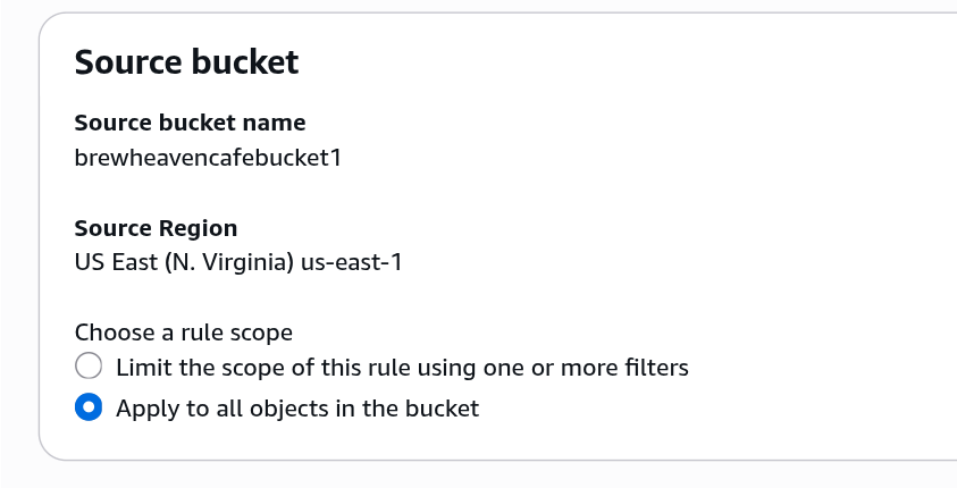
7. On *Replication rules* section select *Create replication rule*.



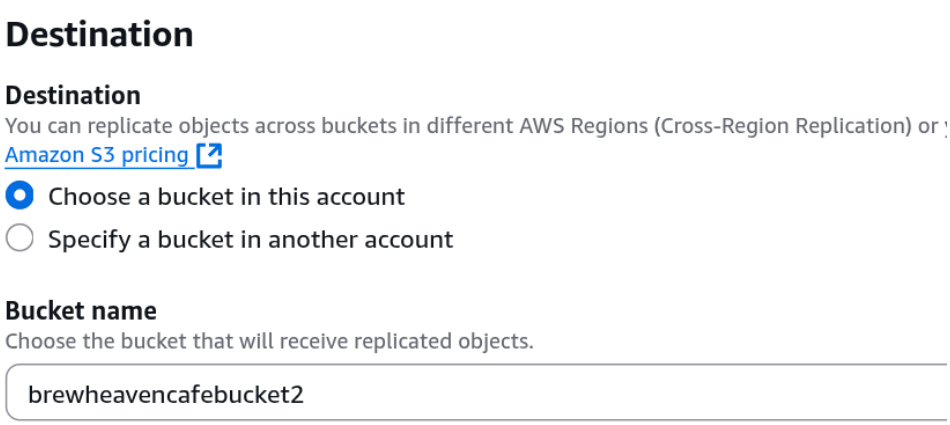
8. Enter the replication rule name.



9. In source bucket section select *Apply to all objects in the bucket*.



10. Choose recently created second bucket as destination.

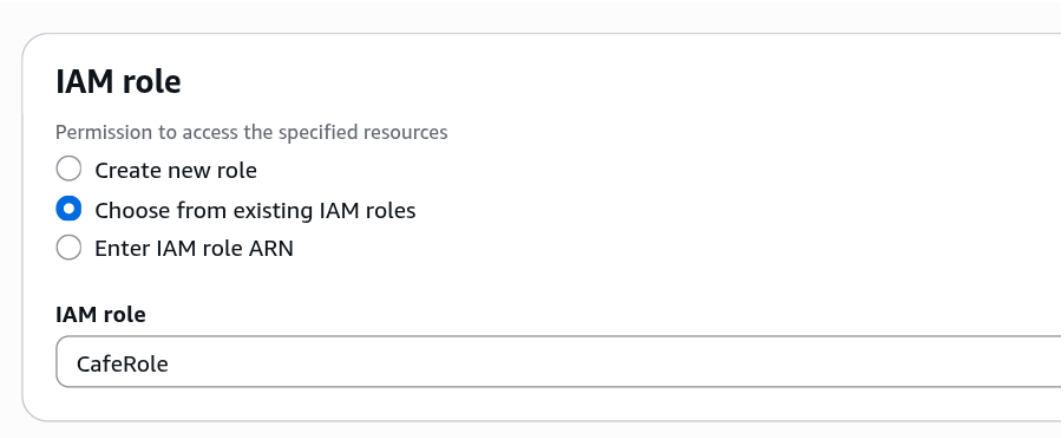


11. Create IAM Role called CafeRole with this policy attached.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:GetObjectVersion",
        "s3:ReplicateObject",
        "s3:ReplicateDelete",
        "s3:ReplicateTags",
        "s3:GetObjectVersionTagging"
      ],
      "Resource": [
        "arn:aws:s3:::SOURCE-BUCKET-NAME",
        "arn:aws:s3:::SOURCE-BUCKET-NAME/*"
      ]
    },
    {
      "Effect": "Allow",
      "Action": [
        "s3:ReplicateObject",
        "s3:ReplicateDelete",
        "s3:ReplicateTags",
        "s3:GetObjectVersionTagging"
      ],
      "Resource": "arn:aws:s3:::DESTINATION-BUCKET-NAME/*"
    }
  ]
}
```

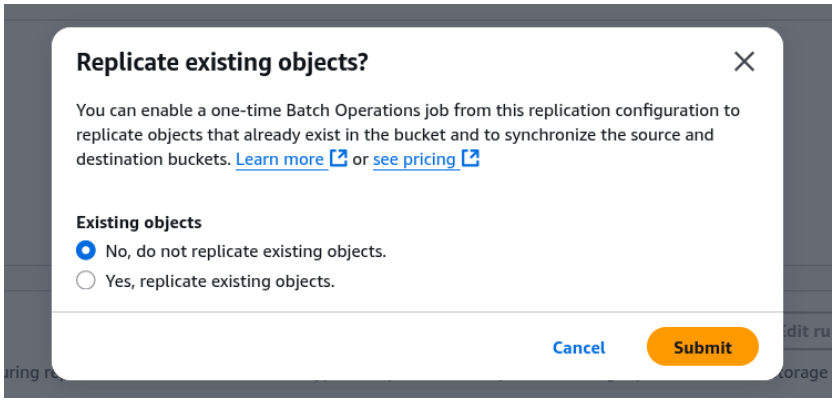
Unfortunately, I am unable to show creation of IAM Role.

12. Select *CafeRole* in the *IAM role* section.

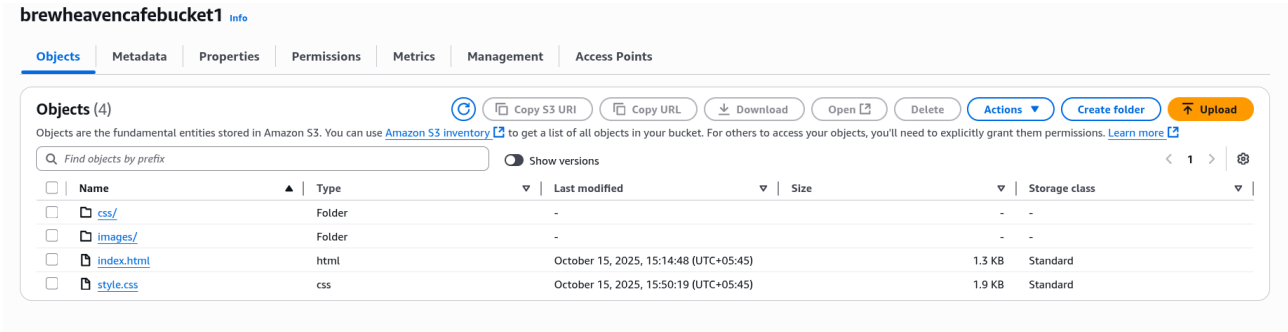


13. Keeping all as it is, proceed with save at the bottom.

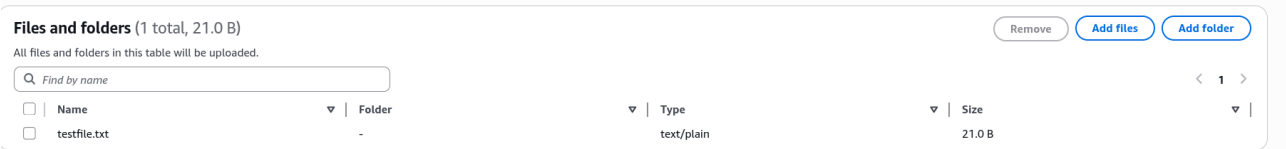
14. If the *Replicate existing objects* dialogue appears select *no*.



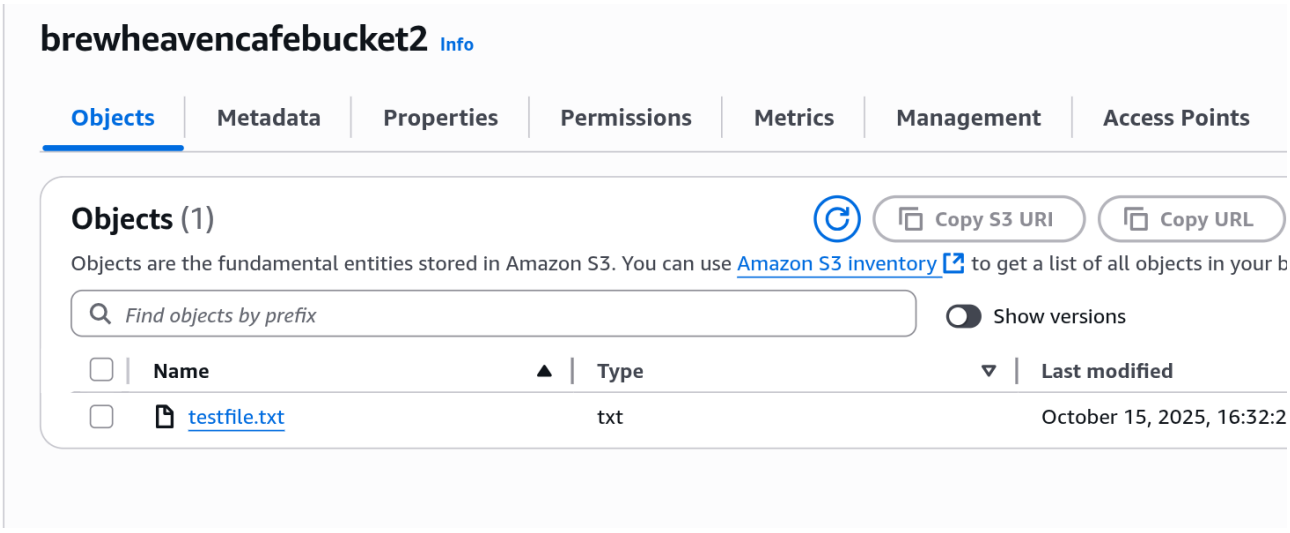
15. To test, open the first bucket and select *Upload* to upload file.



16. Add file and uplaod it.



18. Once uploaded open the second bucket. It may take a while.



Congratulations ! Your file is successfully replicated.